Abdoul Ndiongue

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Education

Carnegie Mellon University — Pittsburgh, PA
BS in Computer Science & Artificial Intelligence — *Expected Graduation: December 2027*
Relevant Coursework: Parallel & Sequential Data Structures and Algorithms (15-210), Computer Systems (15-213), Imperative Programming (15-122), Machine Learning (10-301), Generative AI (10-423), Autonomous Agents (15-482)
Scholarships & Awards: Ron Brown Rales Scholar, Carnegie Mellon Tartan Scholar

Experience

Student Researcher — May 2026 – Present
Machine Learning Department – Carnegie Mellon University — *Pittsburgh, PA*

- **Designed and implemented an autonomous computer-use agent framework** for low-parameter vision-language models, building modular perception, planning, action-selection, and evaluation components for pixel-only desktop interaction tasks.
- **Built an experiment harness for OSWorld-style agent evaluation**, automating task execution, trajectory logging, failure-mode tagging, and reproducible benchmark runs to analyze model behavior across UI navigation, grounding, and planning tasks.
- **Optimized inference and control loops for edge-deployable VLM agents**, using lightweight prompting, action abstraction, observation filtering, and structured error recovery to reduce dependence on frontier-scale cloud models.

Software Engineer Intern — May 2025 – August 2025
Microsoft – Xbox, PC and Console Engagement Team — *Redmond, WA*

- **Engineered a churn-prediction system** using Gradient Boosted Decision Trees (XGBoost) to analyze 15M+ user sessions, identifying high-risk churn patterns with a projected retention impact of 1.8M users annually.
- **Developed and integrated a real-time inference engine** into the Xbox PC app using React and Python, enabling the dynamic delivery of personalized engagement content based on live user behavior metrics.
- **Optimized model performance** by performing extensive feature selection and hyperparameter tuning, achieving an approximate 4.7% improvement in churn reduction metrics through rigorous A/B testing and simulation.

Software Engineer Intern — May 2024 – August 2024
Success Academy Charter Schools — *Manhattan, NY*

- **Identified a manual data-workflow bottleneck and built an automated ingestion service**, eliminating 100+ hours of manual engineering work annually.
- **Engineered a spatial analytics tool implementing a convex hull algorithm** over geospatial datasets to power a real estate site-selection pipeline, improving expansion efficiency by 35%.
- **Designed, built, and deployed 6 internal tools on AWS infrastructure** (S3, Redshift), collaborating with data science and business teams to ship data-driven features 20% faster.

Traceback AI — March 2026
Solo Developer | 2nd Place, Felicis Hackathon (50+ Teams) | Repository Link — *Pittsburgh, PA*

- Engineered an end-to-end system that **intercepts real-time production error signals** (Sentry) to autonomously script, simulate, and verify code fixes in an isolated headless environment, cutting MTTR from hours to minutes.
- **Multi-Agent Orchestration:** Built a tool-loop using Claude 3.5 Sonnet for repository traversal and root-cause analysis, with a three-stage heuristic auditor that verifies AST-parsed patches against generated "Proof of Bug" tests.
- **Built a reactive event-driven architecture** bridging monitoring and deployment, automatically generating and delivering verified Pull Requests to preview environments (Vercel) for human-in-the-loop approval.

Technical Skills

Languages: Python, C, C++, TypeScript, JavaScript, SQL, HTML/CSS
Frameworks & Libraries: React, Node.js, Express.js, FastAPI, Flask, PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, LangChain
Tools & Infrastructure: AWS, Azure, Docker, Git, GitHub, CI/CD, Linux/Unix, Bash, Vercel, Sentry, PostgreSQL, MongoDB, Redshift, S3
Core Skills: REST APIs, Event-Driven Systems, Full-Stack Development, Backend Services, Distributed Systems, Testing, Debugging, Performance Optimization, Data Structures & Algorithms, Machine Learning

Leadership

Interfraternity Council — November 2025 – Present
President — *Pittsburgh, PA*

- **Led executive governance for CMU's Interfraternity Council**, overseeing 8 chapters.
- **Developed risk-management and incident-response processes** to improve consistency, accountability, and communication during conduct cases, chapter concerns, and high-pressure organizational decisions.
- **Served as primary liaison between chapter leaders, university administration, and external stakeholders**, aligning student advocacy with institutional standards, safety expectations, and long-term community interests.